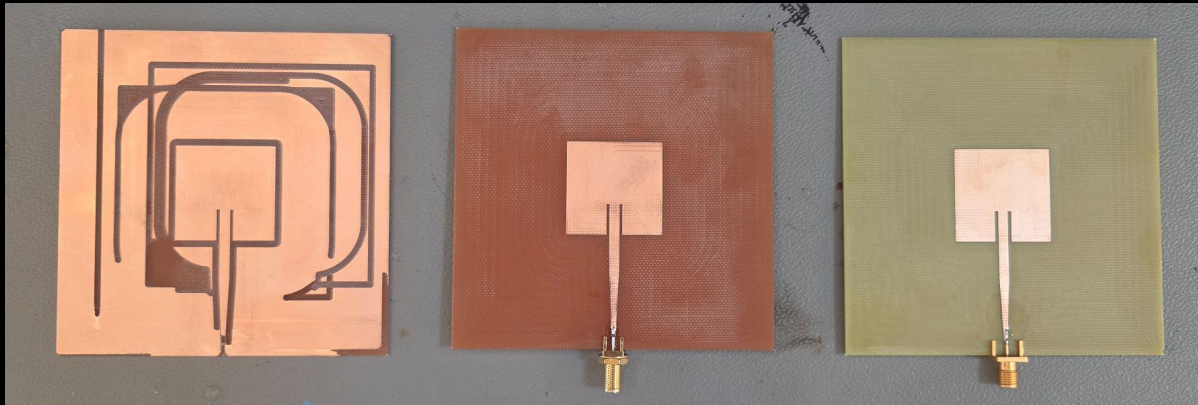
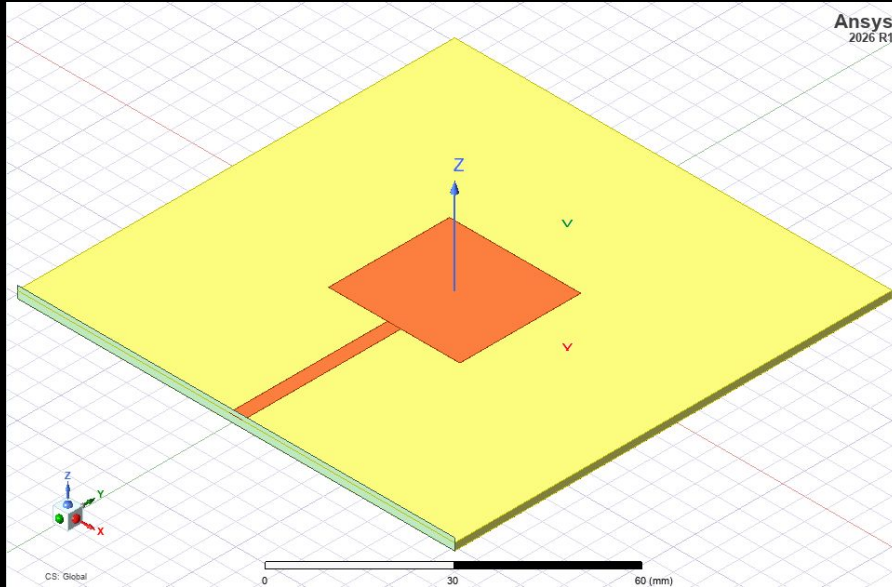


2.44GHz Patch Antenna Design and Testing

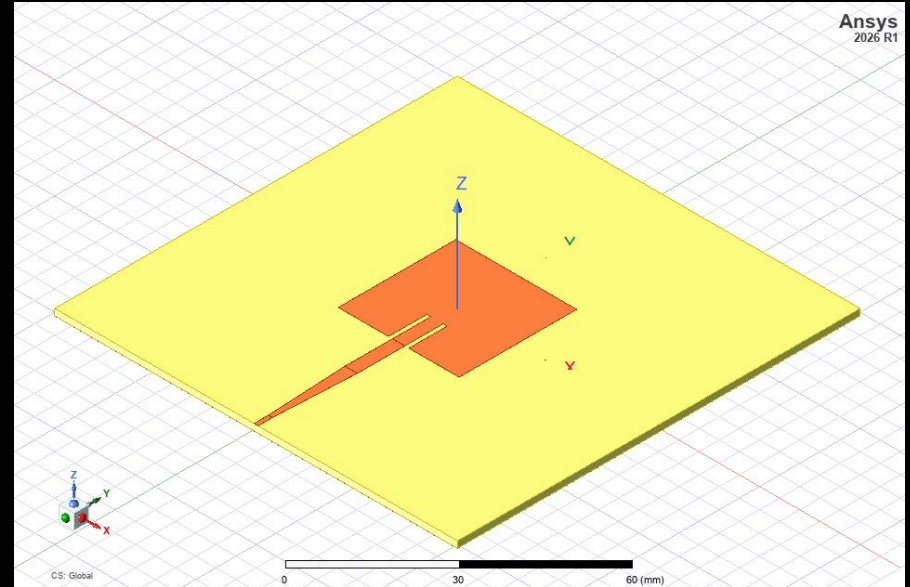
George Crawford-Eng, gcrawfordeng3@gatech.edu



CAD and Design

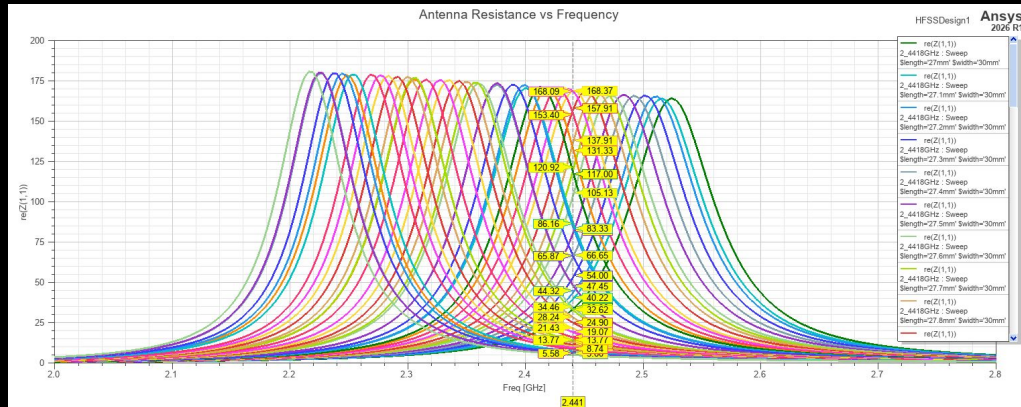
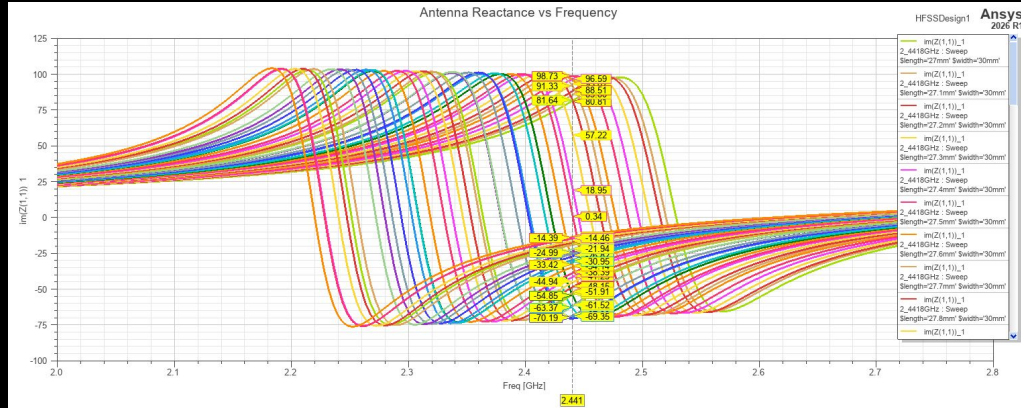


Initial rectangular patch in Ansys
resonant at 2.44GHz



Patch with inset feed and linear
taper for matching

Parameter Sweep



- Length was initially set to half wavelength in FR4
- Actual wavelength is different because E-field is not entirely contained within substrate
- The input reactance (top) & resistance (bottom) was measured across a parameter sweep of the length
- The length was adjusted so that antenna was resonant at 2.44GHz ($\text{Im}(Z_{\text{in}})=0$)

Optimization

Post Analysis Display

Optimization Setup 1 

Result Profile

HFSS version 2026.1.0 started on ECE-2027-040 at 09-08-2026 11:08:51

Variation	\$feed_depth	\$feed_gap_width	\$feed_width	\$height	\$length	\$width	\$feed_width	\$width	Start	Stop	Elapsed	Analysis Machine
1	10mm	1mm	3.062mm	0.017...	29.2mm	30mm	3.062mm	3.06...	11:08:...	11:09:...	00:00:...	Local Machine
2	10.211995...	1mm	3.062mm	0.017...	29.2mm	30mm	3.062mm	3.06...	11:09:...	11:11:...	00:02:...	Local Machine
3	10mm	1mm	3.062mm	0.017...	29.25...	30mm	3.062mm	3.06...	11:11:...	11:13:...	00:02:...	Local Machine
4	9.7880048...	1mm	3.062mm	0.017...	29.2mm	30mm	3.062mm	3.06...	11:13:...	11:16:...	00:02:...	Local Machine
5	10mm	1mm	3.062mm	0.017...	29.14...	30mm	3.062mm	3.06...	11:16:...	11:18:...	00:02:...	Local Machine
6	9.5938864...	1mm	3.062mm	0.017...	29.15...	30mm	3.062mm	3.06...	11:18:...	11:21:...	00:02:...	Local Machine
7	9.7411935...	1mm	3.062mm	0.017...	29.15...	30mm	3.062mm	3.06...	11:21:...	11:23:...	00:02:...	Local Machine
8	9.5938864...	1mm	3.062mm	0.017...	29.18...	30mm	3.062mm	3.06...	11:23:...	11:25:...	00:02:...	Local Machine
9	9.4465793...	1mm	3.062mm	0.017...	29.15...	30mm	3.062mm	3.06...	11:25:...	11:28:...	00:02:...	Local Machine
10	9.5938864...	1mm	3.062mm	0.017...	29.12...	30mm	3.062mm	3.06...	11:28:...	11:30:...	00:02:...	Local Machine
11	9.5312059...	1mm	3.062mm	0.017...	29.16...	30mm	3.062mm	3.06...	11:30:...	11:32:...	00:02:...	Local Machine

Export...

Solver Profile...

Finished at 09-08-2026 11:32:31 (Total Time : 00:23:40)
Solved (below minimum step width, acceptable cost limit NOT satisfied)

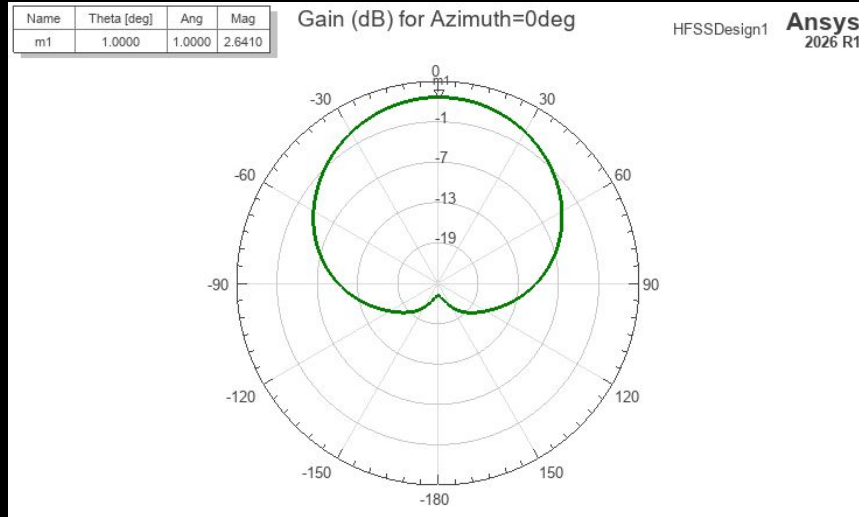
Close

- Once antenna resonance was set, inset feed was added to match input impedance
- Initial feed depth was estimated based on edge resistance
- Feed width was set based on process tolerance
- Gradient based optimizer was used to minimize s11 and set input impedance to 50Ω by altering feed depth and antenna length

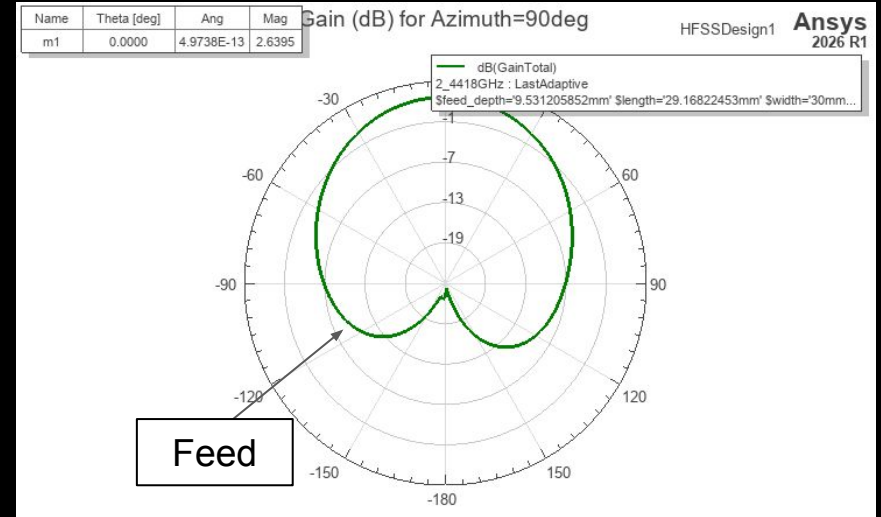
(Simulated S_{11} for antenna is shown to be -22.54 dB and BW is 1.8% BW)



Antenna Characteristics (Cont.)

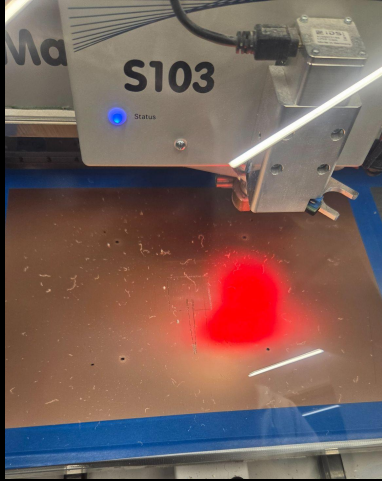


Simulated gain in elevation plane perpendicular to feed. Peak gain is about 2.64dB

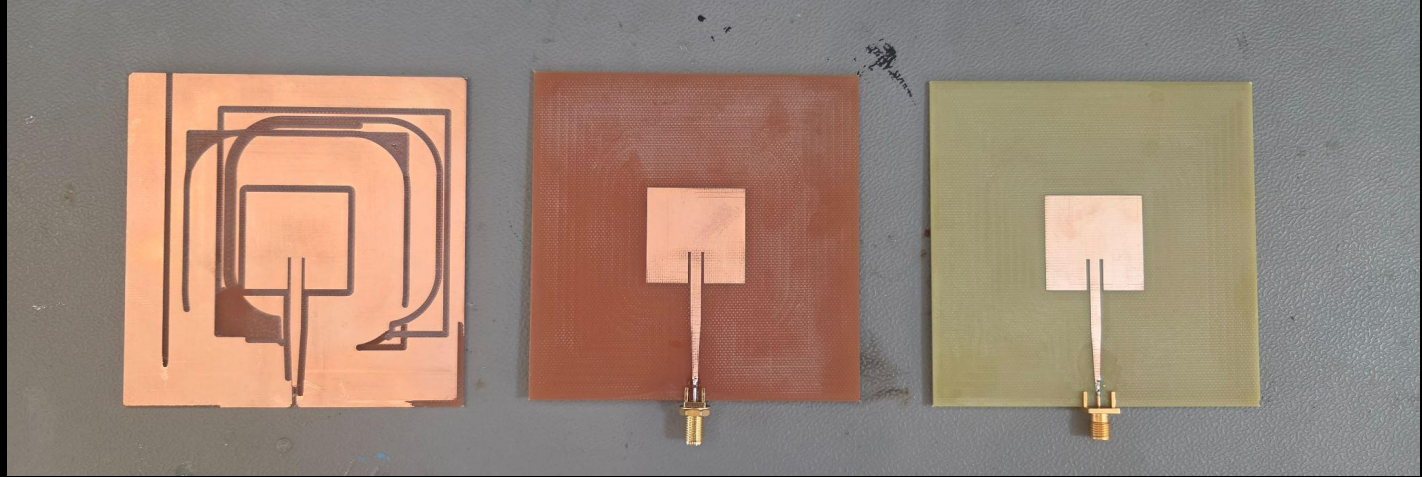


Simulated gain in elevation plane in line with feed. Grating lobe around -180°

Fabrication

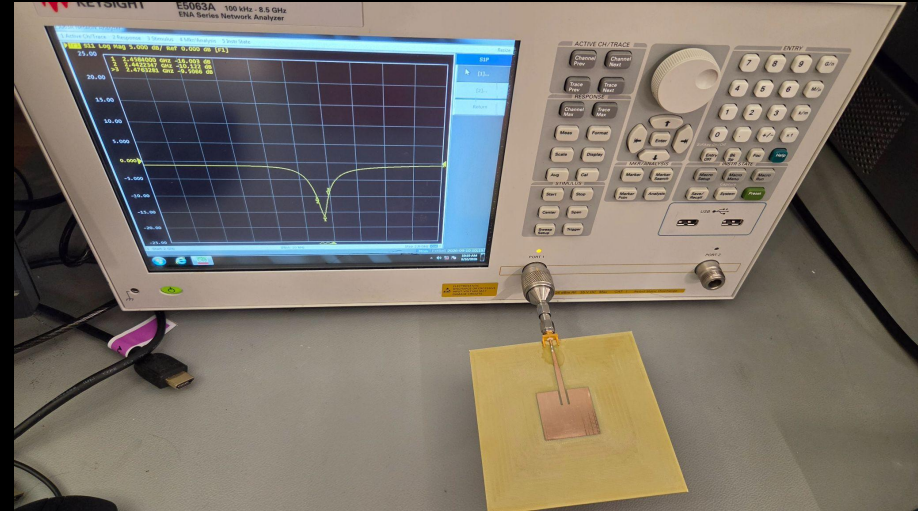
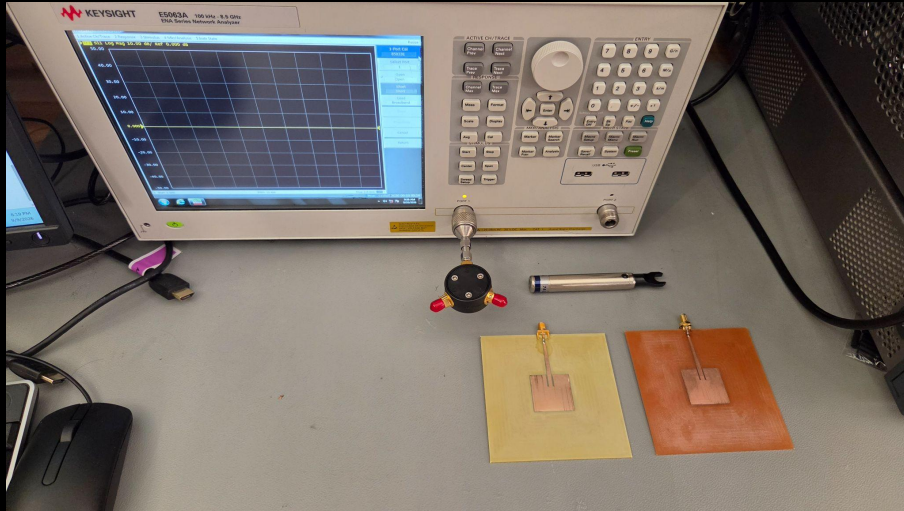


PCB being
fabbed on mill.



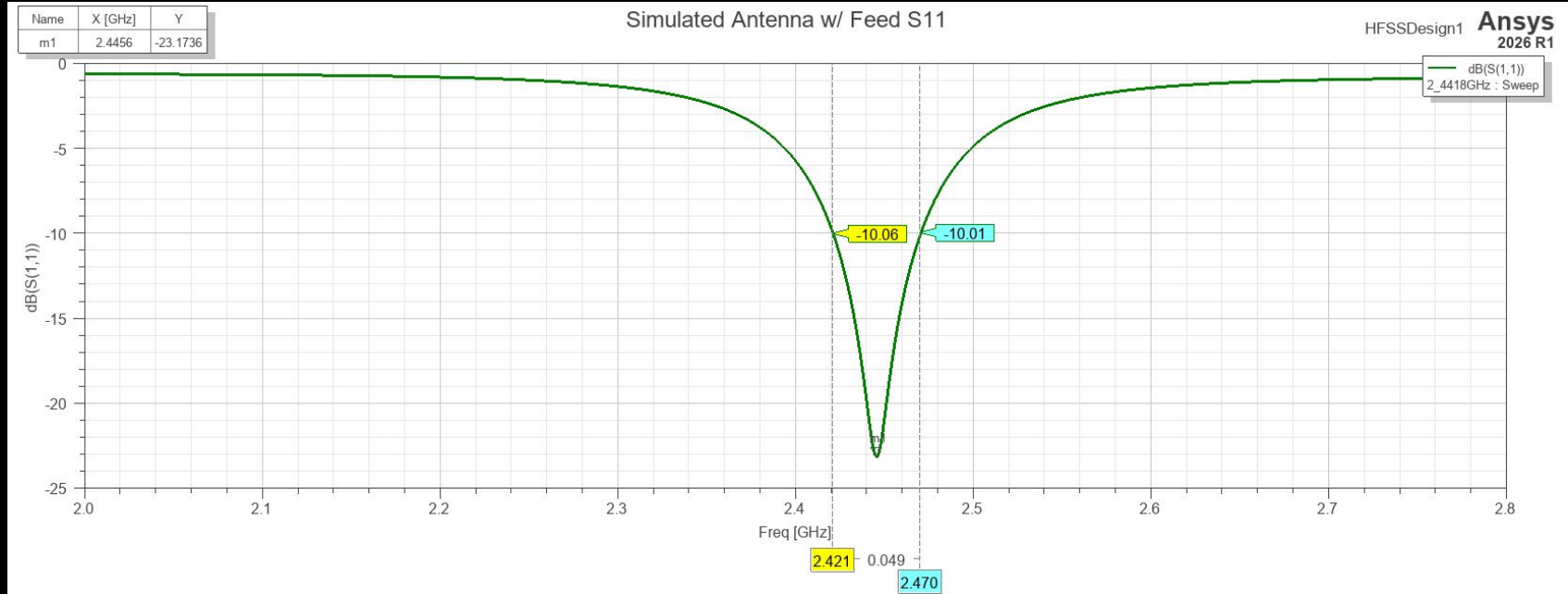
From Left to Right: improper X & Y calibration caused misaligned toolpath. Incorrect z-offset caused uneven substrate height between tools. Final PCB

Calibration and Testing



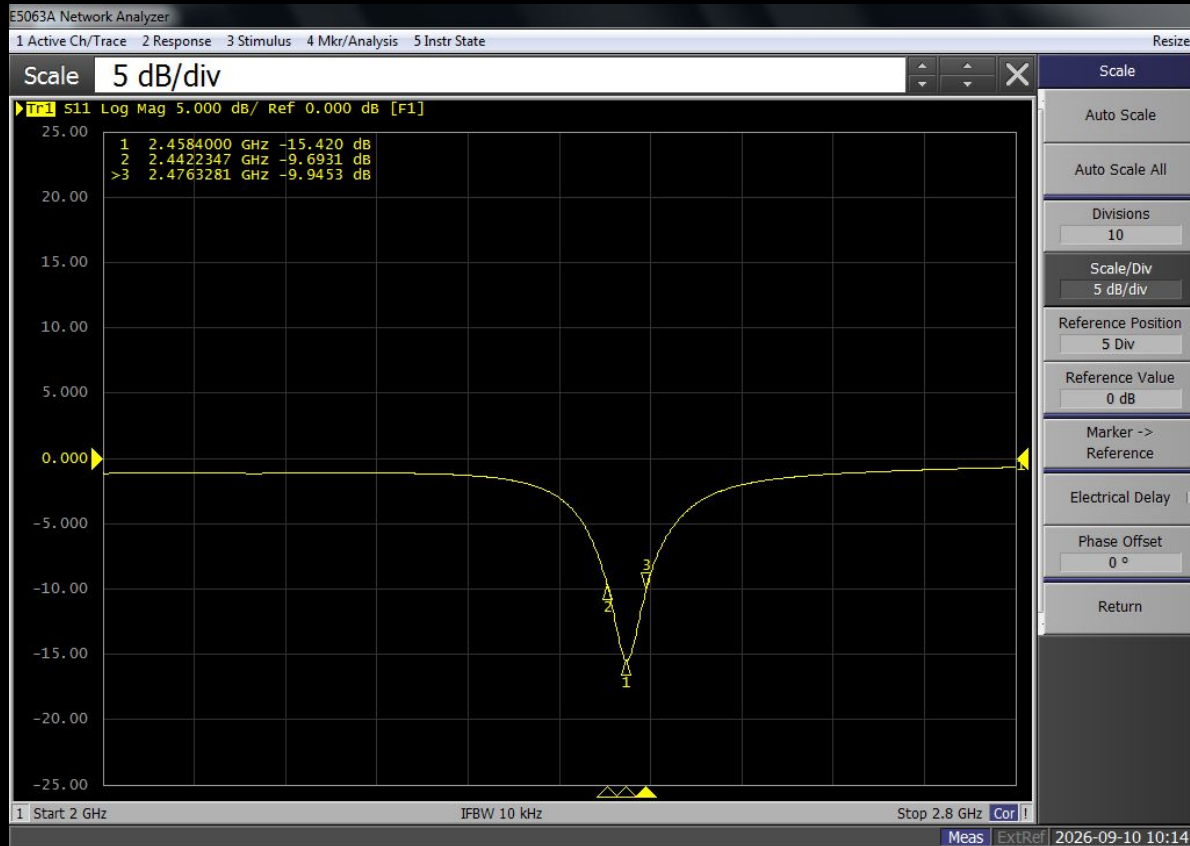
VNA was calibrated using SOL kit to measure S_{11} and input impedance of antenna

Simulated S₁₁



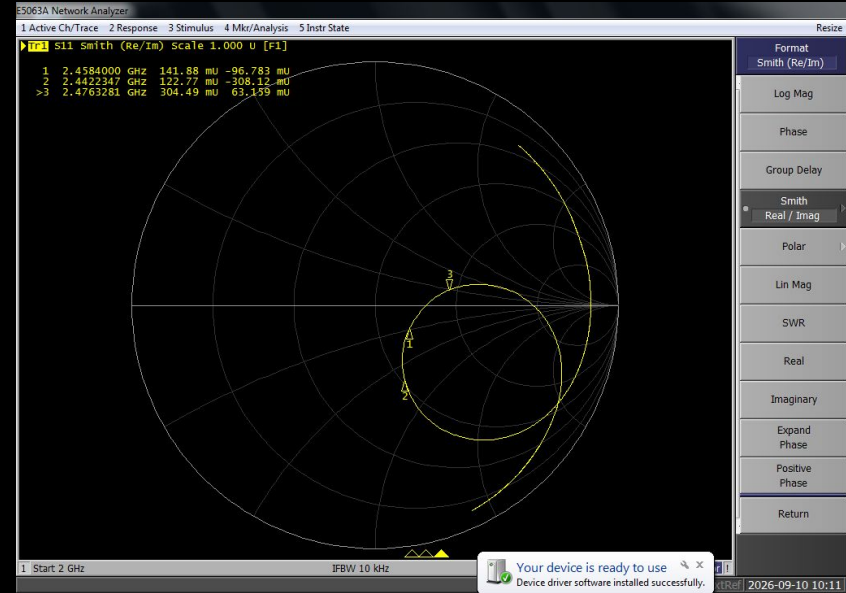
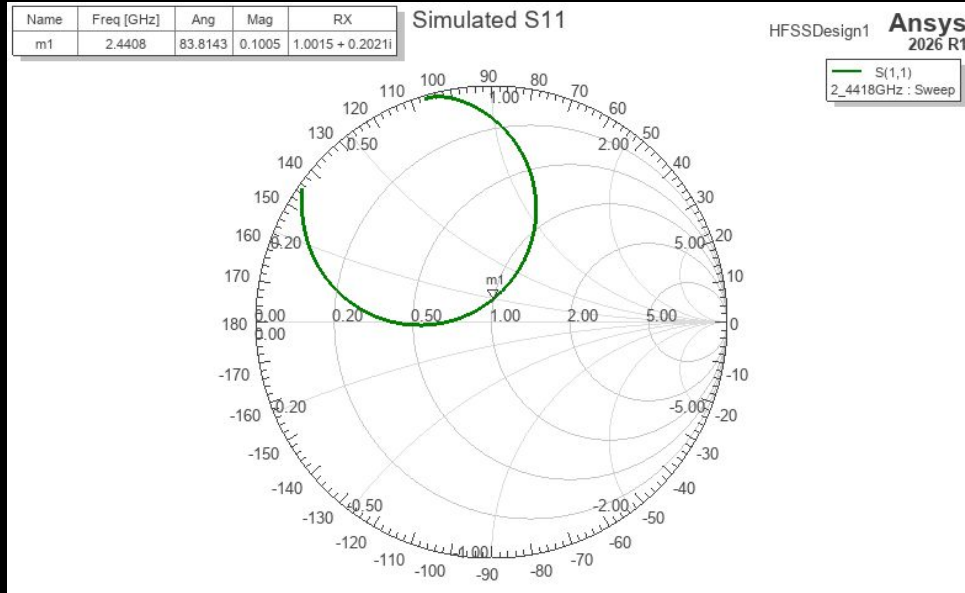
Minimum S₁₁ is -23.17dB at 2.45GHz with 1.8% BW

Measured S11



Minimum S_{11} at 2.46GHz
is -15.42dB with 1.4%

Measured vs Tested Input Impedance



S_{11} plotted on smith chart. Measured is rotated by $\lambda/4$, which may be do to the SMA adaptor between antenna and VNA. From S_{11} the simulated input impedance is $60.1 + j2.1 \Omega$ while the measured was $50 + j0.1 \Omega$

Next Steps

- Measure gain using far-field turntable setup
- Adjust simulation permittivity based on experimental data
- Switch to RT 5880 or lower permittivity substrate to increase BW and Gain to meet bluetooth spec
- Explore alternate topologies such as adding a slit or inverted F design